



Cloud deployment strategies

Now that you're familiar with the different cloud deployment models, let's explore how they shape business strategy. Every organization has unique needs, and selecting the right deployment approach can help optimize infrastructure, meet regulatory requirements, and secure a competitive advantage. In this video, you'll explore the definitions of private, hybrid, and multicloud, and when an organization might choose each approach.

Click play to start the video.



Private cloud, hybrid cloud, and multi-cloud strategies

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Transcript

It's not always possible, or necessary, for an organization to rely solely on the cloud. For example, requirements might call for on-premises infrastructure to work with public cloud services provided by companies, like Google Cloud. With the availability of different cloud options and configurations, it's important to understand what each means.

Let's explore the definitions of private, hybrid, and multicloud, and when an organization might choose each approach. Let's begin with private cloud, which is when an organization has virtualized servers in its own data centers, or those of a private cloud provider, to create its own private dedicated environment.

On-premises servers are also often referred to as private clouds, but generally the distinction can be made that on-premises software runs in a local environment, whereas private cloud is accessed through the internet. Private cloud computing gives an organization many of the benefits of a public cloud — including self-service, scalability, and elasticity — with more customization available than from dedicated on-premises infrastructure. This approach is often used when an organization has already made significant infrastructure investments, or if, for regulatory reasons, data must be kept on-premises.

In contrast, a hybrid cloud is one in which applications are running in a combination of different environments. The most common example is combining a private and public cloud environment, like an on-premises data center, and a public cloud computing environment like Google Cloud.

Finally, there's multicloud, which describes architectures that combine at least two public cloud providers, such as Google Cloud, Amazon Web Services, Microsoft Azure, or others. An organization might choose multicloud if they want to take advantage of the key strengths of different public cloud providers. Organizations may also operate a combination of on-premises and multiple public cloud environments, effectively being both hybrid and multicloud simultaneously. A hybrid cloud approach is one of the most common infrastructure setups today because organizations can continue to use their on-premises servers while also taking advantage of public cloud.

Hybrid and multicloud strategies

As explained in the video, hybrid and multicloud are often used interchangeably, but there is a distinct difference. Here's a breakdown of exactly what distinguishes the two.



Hybrid cloud

Hybrid cloud describes applications that run in a combination of different environments. The most common hybrid cloud example is combining a public and private cloud environment, like an on-premises data center, and a public cloud computing environment, like Google Cloud.



Multicloud

Multicloud architectures combine at least two public cloud providers. Organizations might operate a combination of on-premises and multiple public cloud environments, therefore implementing both hybrid and multicloud simultaneously.

What is a hybrid or multicloud strategy used for?

Let's explore some different business requirements, drivers, and use cases that lead an organization to choose a hybrid or multicloud strategy.

To learn more, expand each of the eight items.

Access to the latest technologies.



Running workloads in multiple clouds empowers organizations to leverage the latest innovations and capabilities from each cloud provider, thus taking a best-in-class approach to cloud features and obtaining the scale, security, and agility to innovate fast. Cloud can help organizations build out capabilities, such as advanced analytics services, that might be difficult, or impossible, to implement in existing environments.

Modernize at the right pace.



With a hybrid cloud, organizations can migrate applications to the cloud at the pace that makes sense for their business and transform their technical infrastructure over time.

Improved return on investment.



By adding a public cloud provider to their existing on-premises infrastructure, organizations can expand their cloud computing capacity without increasing their data center expenses. This can help reduce CapEx or general IT spending, and improve transparency regarding costs and resource consumption.

Flexibility through choice of tools.



Hybrid and multi-cloud strategies have advantages for organizations as a whole, but specifically benefit development teams that are working on different projects and tackling unique challenges across different lines of business. A wider choice of tools and developer talent can be applied to a particular business problem, which means responding better to changing market demands. It also avoids vendor lock-in concerns.

Improve reliability and resiliency.



Organizations can distribute core workloads across multiple cloud and on-premises infrastructures to reduce downtime and concerns about over-dependence on a single source of failure. This approach can improve the quality and availability of a service.

Maintain regulatory compliance.



Many industries have rules from governmental or regulatory bodies regarding where their app can operate. Adopting a hybrid solution is an effective way for an organization to ensure compliance with regional data governance, residency, or digital sovereignty requirements.

Running apps on-premises.



Organizations may have regulated applications that must remain on-premises or mainframe systems that are difficult to move to the cloud. A hybrid approach provides the freedom to innovate while still

meeting legacy technology needs.

Running apps at remote edge locations.



Organizations in industries that run distributed apps at remote locations, such as kiosks in retail or networks in telecom, can benefit from hybrid cloud. These apps often require improved performance and low latency, and a hybrid approach lets them run select apps at the network edge.

Open standard vs. open source

Two elements that are fundamental to hybrid and multi-cloud strategies are **open standard** and **open source** technologies. These terms can often be confused, so let's explore these concepts in a little more detail.

Click each card to learn more.

Open standard

Refers to software that follows particular specifications that are openly accessible and usable by anyone. They have guidelines for software functionality, which help avoid vendor lock-in and ensure that

Open source

Refers to software whose source code is publicly accessible and free for anyone to use, modify, and share. A decentralized community generally develops open source software as a public collaboration, based on a

Knowledge check

Why would an organization choose a hybrid or multicloud strategy?

Select all that apply.

To eliminate the need for any on-premises IT staff or legacy applications.



To improve return on investment (ROI) by expanding computing capacity without increasing data center capital expenses (CapEx).



To avoid vendor lock-in concerns and take advantage of the latest, best-in-class technologies and capabilities from different cloud providers.



To improve reliability and resiliency by distributing core workloads across multiple infrastructures.

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